

1.1 -Technical specifications

General

	Reference	Technical specifications
Motorcycle dimensions	Total length	2120 mm
	Total width	784 mm
	Total height	1088 mm
	Wheelbase	1450 mm
	Height of handlebar grip from ground	990 mm
	Seat height	810 mm
	Front footrest height	360 mm
	Rear footrest height	531 mm
	Minimum ground clearance	196 mm
Frame	Type	Chromium-molybdenum tubular trellis frame
Subframe	Type	Cast aluminium
	Steering head angle	24°
	Steering angle	32°
	Trail	87 mm
	Front wheel	Light alloy with five Y-shaped spokes
	Front suspension (1100 ABS)	SHOWA: Upside-down hydraulic fork with adjustable compression, rebound and spring preload. Fork legs Ø 43 mm
	Front suspension (1100 S ABS)	ÖHLINS: Upside-down hydraulic fork with adjustable compression, rebound and spring preload. Fork legs Ø 43 mm
	Travel along leg axis	130 mm (1100 ABS - 1100 S ABS)
	Rear suspension (1100 ABS -1100 S ABS)	SACHS (1100 ABS) ÖHLINS (1100 S ABS) The rear suspension system uses a hydraulic monoshock absorber with rebound and spring preload adjustment. Rear wheel travel: 148 mm. Shock absorber stroke: 59.5 mm.
	Front wheel size	MT 3.50x17"
	Front tyre size	120/70 - ZR 17
	Rear wheel	Light alloy with five Y-shaped spokes
	Rear wheel size	MT 5.50x17"

	Rear tyre size	180/55 - ZR 17
	Type of tyre	Radial tubeless tyres
	Front brake	Hydraulic with 2 callipers
	Rear brake	Hydraulic with 1 calliper
Engine	Type	Longitudinal 90° "L" twin cylinder, four-stroke.
	Bore	98 mm
	Stroke	71.5 mm
	Displacement	1078 cm ³
	Compression ratio	10.7±0.5:1
	Timing system	With toothed belt, single overhead camshaft, 2 valves per cylinder and 4 rocker arms.
	Lubrication system	Forced lubrication by pump, with oil cooler
	Oil pump type	Gear pump with bypass valve
	Cooling system	Air-cooled
	Air filter	Filter element in the airbox
	Crankshaft type	One-piece

Colours

	Description	Code no
1100 ABS	Ducati Anniversary Red	473.101 (PPG)
	Transparent	228.880 (PPG)
	Red frame with grey wheels	
	Matt Silver	928D998 (PAL)
	Clear lacquer	923i1281 (PAL)
	Red frame with Matt Silver wheels	
	Ducati gloss black	248.514 (PPG)
	Clear lacquer	228.880 (PPG)
	Matt black frame with Matt Silver wheels	
1100S ABS	Ducati Anniversary Red	473.101 (PPG)
	Transparent	228.880 (PPG)
	Red frame with golden wheels	
	Pearl white primer	490.019 (PPG)
	Enamel	*0040 (PPG);
	Clear lacquer	228.880 (PPG)
	Matt black frame and golden wheels	

Transmission

	Reference	Technical specifications
	Clutch	Multiple disc dry clutch
	Clutch control	Hydraulic
	Gearbox	6 speed
	Primary drive	32/59
	Transmission ratio	1.844
	Final drive	15/39
	Gearbox type	With constant mesh spur gears, operated by a lever on the left side of the motorcycle

Gear ratios		
1 st	15/37	
2 nd	17/30	
3 rd	20/27	
4 th	22/24	
5 th	24/23	
6 th	28/24	

Timing system/valves

	Reference	Adjusting clearance	Checking clearance
Timing diagram	With 1 mm valve clearance		
	Intake	Opening 15° B.T.D.C. Closing 65° A.B.D.C.	
	Exhaust	Opening 62° B.B.D.C. Closing 19° A.T.D.C.	
	Intake valve diameter	45 mm	
	Exhaust valve diameter	40 mm	
Valve lift	With 0 mm valve clearance	Intake 11.20 mm Exhaust 10.78 mm	
	Opening rocker arm - intake	0.10 to 0.15 mm	0.05 to 0.15 mm
	Opening rocker arm - exhaust	0.10 to 0.15 mm	0.05 to 0.15 mm
	Closing rocker arm - intake	0 to 0.05 mm	0 to 0.20 mm
	Closing rocker arm - exhaust	0 to 0.05 mm	0 to 0.20 mm
Timing belt tensioning	DDS	140±5 Hz (horizontal) 140±5 Hz (vertical)	140±5 Hz (horizontal) 140±5 Hz (vertical)
	Minimum limit value	70 Hz	70 Hz

Cylinder/Piston

	Reference	Standard value	Service limit
Cylinder barrel	Max. bore ovality		0.03 mm
	Max. bore taper		0.03 mm
	Nominal diameter	98 mm	
	Size group A	98.000 to 98.010 mm	
	Size group B	98.010 to 98.020 mm	
	Size group C	98.020 to 98.030 mm	
Piston-to-cylinder bore clearance		0.025 to 0.045 mm	
Piston	Nominal diameter	98 mm	
	Size group A	97.965 to 97.975 mm	
	Size group B	97.975 to 97.985 mm	
	Size group C	97.985 to 97.995 mm	
Connecting rod	Connecting-rod big-end diameter	43.67 mm	
	Crankshaft class	Connecting rod class	Bearings colour
Big-end bearing pairings	A	A	Blue + Yellow
	B	A	Yellow + Yellow
	A	B	Blue + Blue
	B	B	Blue + Yellow
Big-end bearing-to-crankpin clearance		Ø 40.017 mm/0 to 0.016 mm	

Gudgeon pin-to-piston clearance		0.015 to 0.058 mm	
	Nominal diameters	Ø 18 mm	
	Piston	Ø 18 mm/0.020 to 0.015	
	Gudgeon pin	Ø 18 mm/0 to 0.004	
Gudgeon pin-to-connecting rod clearance		0.006 to 0.028 mm	
Cylinder compression	Standard Minimum Maximum permissible difference	11 to 13 bar	min. 10 bar max. 2 bar

Gearbox

	Reference	Standard value	Service limit
Gearbox shafts	End float		0.05 to 0.020 mm
Selector drum	End float		0.10 to 0.040 mm
Gear selector fork	Selector fork end thickness	3.90 to 4.00 mm	
	Selector fork-to-gear clearance	0.070 to 0.285 mm	0.4 mm

Cooling system

	Reference	Technical specifications
Type	Air-cooled: heat dispersion through the extensive finning on the two cylinder barrel/head assemblies	

Front wheel

	Reference	Standard value	Service limit
Minimum tread depth	In the most worn part		2 mm
Tyre pressure	Cold	2.25 bar	
Wheel shaft runout	On 100 mm		0.05 mm
Wheel rim runout	Radial		1 mm (0.04 in.)
	Lateral		0.5 mm (0.02 in.)
Air gap	Range	1.3 to 1.9 mm	

Front suspension (1100 ABS)

	Reference	Technical specifications
Type	SHOWA: Upside-down hydraulic fork with adjustable compression, rebound and spring preload, 43 mm Ø.	
	Travel along leg axis	130 mm
Fork	Hydraulic damping standard setting. Undo adjusters from the fully closed position (turn counter clockwise).	Compression: 1.5 turns ±1/4 turn Rebound: 1.5 turns ±1/4 turn
	Spring preload	22 mm (3 turns from fully open position)
	Oil charge per leg	524±2.5 cm ³ (per leg)

Front suspension (1100 S ABS)

	Reference	Technical specifications
Type	ÖHLINS:	

Upside-down hydraulic fork with adjustable compression, rebound and spring preload, 43 mm Ø.

Fork	Travel along leg axis	130 mm
	Hydraulic damping standard setting. Undo adjusters from the fully closed position (turn counter clockwise).	Compression: 8 clicks Rebound: 12 clicks
	Spring preload	22 mm (3 turns from fully open position)
	Oil charge per leg	140 mm, measured with no spring, per leg

Rear wheel

	Reference	Standard value	Service limit
Minimum tread depth	In the most worn part		2 mm
Tyre pressure	Cold	2.5 bar	
Swingarm pivot shaft runout	On 100 mm		0.2 mm
Wheel rim runout	Radial		1 mm (0.04 in.)
	Lateral		0.5 mm (0.02 in.)
Drive chain	Make	REGINA	
	Type	525 ZRPK	
	Dimensions	5/8" x 1/4"	
	No. of links	104	

Rear suspension (1100 ABS)

	Reference	Technical specifications
Type		Sachs, progressive, with adjustable rebound and spring preload
Shock absorber	Stroke	59.5 mm
	Hydraulic damping standard setting. Turn the adjusters fully clockwise until they stop, then turn them anticlockwise	Rebound: 13 clicks
	Spring preload	18 mm
	Wheel travel	148 mm

Rear suspension (1100 S ABS)

	Reference	Technical specifications
Type		ÖHLINS, progressive, with adjustable rebound and spring preload
Shock absorber	Stroke	59.5 mm
	Hydraulic damping standard setting. Turn the adjusters fully clockwise until they stop, then turn them anticlockwise	Rebound: 15 clicks
	Spring preload	10 mm
	Wheel travel	148 mm

Hydraulic brakes

	Reference	Standard value	Service limit
FRONT			
Brake disc	Type	Drilled dual disc	
	Shim	4.5 mm	4.0 mm (min.)
	Carrier (flange) material	Steel (1100 ABS)	

		Aluminium (1100 S ABS)
Brake calliper	Braking surface material	Steel
	Diameter	320 mm
	Make	Brembo
	Type	P4.32K - 4 piston
Brake master cylinder	Calliper cylinder diameter	32 mm
	Pad friction material	FERIT I/D 450-FF
	Type	PR 18/19
	Master cylinder diameter	18 mm
REAR		
Brake disc	Type	Drilled disc
	Shim	5 mm 4.5 mm (min.)
	Material	Steel
	Diameter	245 mm
Brake calliper	Make	Brembo
	Type	P34C - 2 pistons
	Calliper cylinder diameter	34 mm
	Pad friction material	FERIT I/D 450 FF.
Brake master cylinder	Type	PS11
	Master cylinder diameter	11 mm

Charging system/alternator

	Reference	Technical specifications
Battery	Voltage	12 V/10 Ah
	Type	Sealed, maintenance free
Generator	Capacity	12 V - 480 W

Ignition system

	Reference	Technical specifications
Ignition	Type	Integrated with Siemens electronic injection
Starting	Type	Electric starter motor, 12 V - 0.7 kW.
Spark plugs	Make and type	CHAMPION RA6HC (or, as an alternative, NGK DCPR8E)
	Electrode gap	0.7 to 0.8 mm

Fuel system

	Make	Type
	Unleaded fuel	95-98 RON
	Throttle body	Ø 45 mm
	Injectors per cylinder	1
	Holes per injector	8

Injection system

	Make	Type
Electronic Control Unit	Siemens	M3C Sinerject

Lights/instrument panel

	Reference	Technical specifications
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Bulbs	High beam	H1 (12 V - 55 W)
	Low beam	H7 (12 V - 55 W)
	Number plate light	C5W (12 V - 5 W)
	Side light	H6 (12 V - 6 W)
	Turn signal light	RY10W (12 V - 10 W)
	Tail light and brake light	LED
Fuses	Regulator	30 A
	Instrument panel	5 A
	Injection	20 A
	Key ON	10 A
	ECU	5 A
	El. item	15 A
	High beam and low beam headlights, parking lights	15 A
	Spare	20 A
	Spare	15 A